

Practice Questions – Normal Distribution

Q1) Given that X is a random variable that is normally distributed with $\mu = 30$ and $\sigma = 4$. Determine the following:

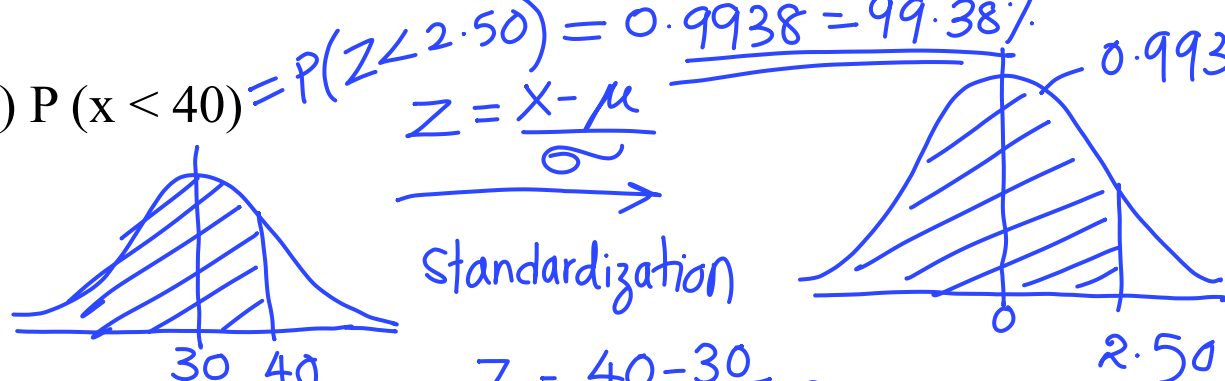
$$X \sim N(\mu = 30, \sigma = 4)$$

1) $P(X < 40) = P(Z < 2.50) = 0.9938 = 99.38\%$

$Z = \frac{X - \mu}{\sigma}$

Standardization

$Z = \frac{40 - 30}{4} = 2.5$

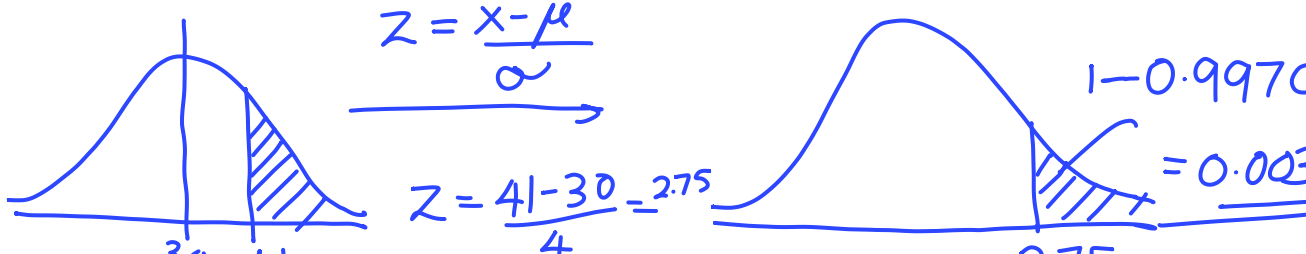


2) $P(X > 41)$

$Z = \frac{X - \mu}{\sigma}$

$Z = \frac{41 - 30}{4} = 2.75$

$1 - 0.9970 = 0.003$

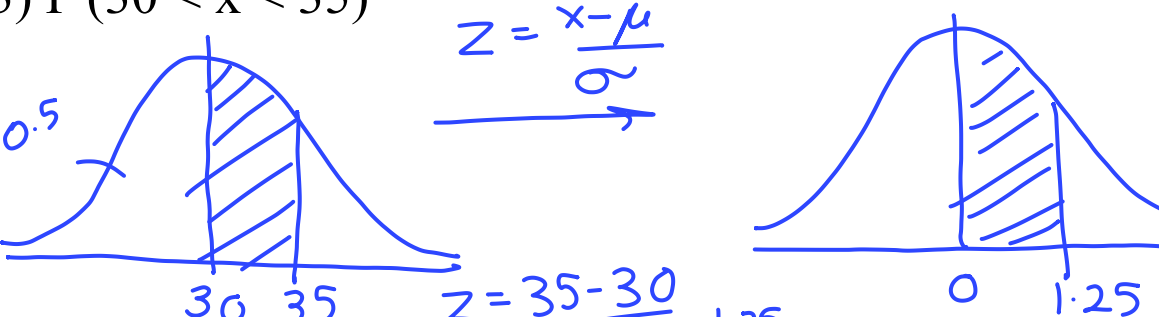


3) $P(30 < X < 35)$

$Z = \frac{X - \mu}{\sigma}$

$Z = \frac{35 - 30}{4} = 1.25$

0.5



4) 95th percentile.

0.95

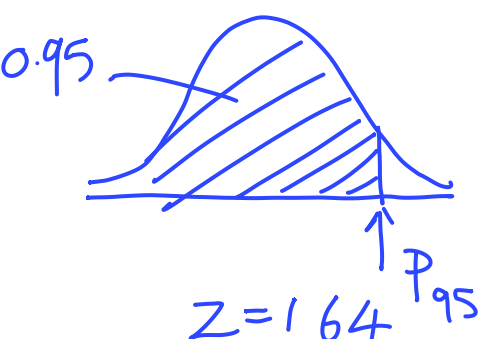
$Z = 1.64$ P_{95}

$Z = \frac{X - \mu}{\sigma}$

$1.64 = \frac{X - 30}{4}$

$X = 36.56$

0.89435
- 0.50000
0.39435

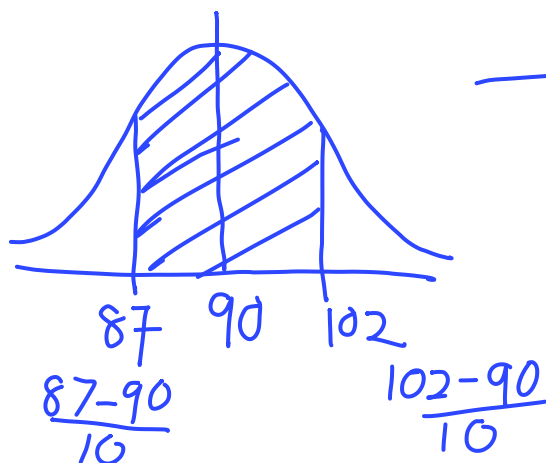


Score $\frac{\text{out of } 800}{800} = 756$, 97^{th} percentile
 $X = 405$, 67^{th}

2) The speeds of buses of a certain brand are normally distributed with a mean of 90 km/hr and a standard deviation of 10 km/hr.

$$X \sim N(\mu = 90, \sigma = 10)$$

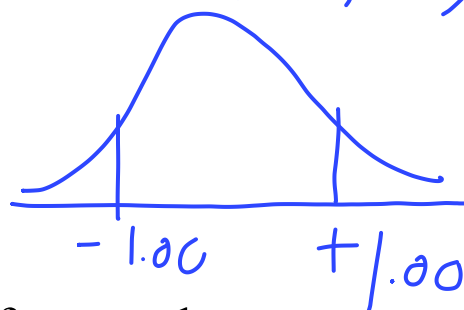
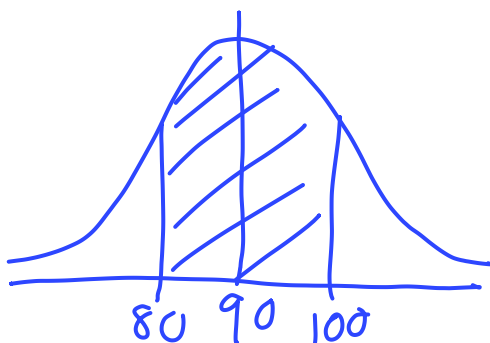
i) What percentage of buses will ride within speed 87 km/hr and 102 km/hr?



ii) What percentage of buses will fall within one standard deviation of the mean speed?

$$\mu \pm 1\sigma \quad 90 \pm 10$$

(80, 100)



iii) Calculate the 90th percentile for speed.

